

Auteur: Rob Willemsen
Studierichting: Informatica
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$$\begin{aligned}f(x) &= \frac{x^2 - x - 2}{x^2 + 5x + 6} \\f'(x) &= \frac{(2x - 1)(x^2 + 5x + 6) - (x^2 - x - 2)(2x + 5)}{(x^2 + 5x + 6)^2} \\&= \frac{(2x^3 + 9x^2 + 7x - 6) - (2x^3 + 3x^2 - 9x - 10)}{(x^2 + 5x + 6)^2} \\&= \frac{2x^3 + 9x^2 + 7x - 6 - 2x^3 - 3x^2 + 9x + 10}{(x^2 + 5x + 6)^2} \\&= \frac{6x^2 + 16x + 4}{(x^2 + 5x + 6)^2}\end{aligned}$$

References